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High-throughput parallelized testing of membrane electrode assemblies for CO_2 reduction†

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High-throughput characterization of electrochemical reactions can accelerate discovery and optimization cycles, and provide the data required for further acceleration *via* machine-learning guided experiment planning. There are a range of high-throughput methods available for catalyst discovery. However, the development and testing of electrochemical systems – integrated electrocatalysts, membranes, and electrodes – currently relies on serial, labor-intensive lab processes. Membrane electrode assembly (MEA) cells have shown particular promise in carbon dioxide (CO_2) reduction, providing commercially viable reaction rates. Experimental testing of MEAs is slow, requiring a serial assembly process that can result in electrode compression levels that are non-uniform over the cell area and challenging to reproduce. Here we demonstrate a new MEA testing system that offers an accelerated, parallelized assembly process and enables high-throughput electrochemical system testing. The approach accelerates electrochemical system testing, controls compression and improves repeatability and reliability. We benchmark our system with CO_2 reduction to ethylene, running 10 MEA experiments in parallel, demonstrating an acceleration factor up to 12× over conventional approaches, and achieving a cell-to-cell gas selectivity deviation of ±2.5%.

Introduction

High-throughput experimentation (HTE), such as rapid serial or highly parallelized processing, can accelerate materials discovery cycles and increase data acquisition rates to enable advanced statistical processing, including those based on machine learning. HTE has been developed extensively for the accelerated discovery of battery electrolytes, pharmaceutical drugs, organic small molecules, inorganic solar harvesting materials and other applications.[1] Mennen *et al.* reported a centralized HTE system capable of designing pharmaceutical reactions by integrating manual, automated, and analytical devices.[2] Another HTE platform was demonstrated by Bédard *et al.*, showing reconfigurability to a variety of chemical reactions, which enabled an accelerated and automated optimization scheme for several experiments.[3] Integration of HTE robotics and machine learning was used in the discovery of new battery electrolytes.[4] In pharmaceuticals,[5] an HTE platform enabled the optimization of nanoscale experiments and minimized the use of precious metals. Catalyst discovery for the carbon dioxide reduction reaction (CO_2RR) is another area in which HTE platforms have enabled acceleration.[6–10]

The field of the electrochemical CO_2RR has advanced greatly over the past decade, improving energy efficiency, stability at industrially relevant current densities, and scalability, in the generation of a variety of products.[11–13] Electrochemical H-cells are commonly employed in CO_2RR experiments for initial catalyst screening. However, these systems are limited to low current densities.[14,15] Flow cells can provide a gaseous CO_2 reactant directly to the electrode, thus enabling higher current densities. However, flow cells suffer electrode flooding issues, and high ohmic losses.[12,16] Membrane electrode assembly (MEA), or 'zero-gap', electrolyzers offer a path to scale, with high stability, high energy efficiency, and practical current densities.[12,17]

In MEA cells, the performance of the CO_2RR depends highly on the local environment as well as the catalyst itself.[11,18] Control of the local environment requires that

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system variables—such as anolyte pH, applied voltage, temperature, and the flow rate of CO_2—and the catalyst be tested together to evaluate the efficiency of the CO_2RR. The complexity of this interplay between the catalyst and system is very challenging to probe with the inherently low-throughput system testing platforms available.[19,20] Conventional experiments with MEA cells employ a tedious cell assembly procedure, manual configuration of testing equipment, and manual data collection. Standard commercially available MEA cells[21] require manual bolting, creating non-uniform compression, damage to the membrane or the electrode, and a lack of repeatability – from experiment-to-experiment and user-to-user. Gas diffusion electrode (GDE) compression is the limiting factor for the mass transport of reactants and products as well as maintaining a triple phase boundary at the active reaction site. These issues have made MEA preparation and testing a main bottleneck in the advancement of electrochemical systems, such as in the discovery of improved catalysts for the CO_2RR.

In this work, we demonstrate a high-throughput testing platform that can accelerate the assembly of MEAs for CO_2RR experiments by greater than 10×. A new MEA design with a more reliable assembly process enables conducting multiple experiments in parallel. Our system accommodates 10 MEA cells, is able to run 10 experiments simultaneously, and further shows enhanced repeatability when testing the same CO_2RR catalyst compared to a conventional MEA. The presented HTE platform overcomes the challenge of slow preparation processes and inconsistent compression force typically experienced with single MEA testing. Our system is also suitable for integration with automated real-time product analysis tools such as selected ion flow tube mass spectrometry (SIFT-MS) and high-throughput electrode synthesis processes, which can dramatically shorten the overall innovation time for new electrocatalyst discovery.

Experimental setup

A typical single MEA test setup for the CO_2RR includes a dedicated potentiostat to deliver current/voltage to the cell, a dedicated electrolyte pump along with a dedicated electrolyte container for the anode side, and a mass flow controller to monitor the CO_2 input. All the typical testing equipment fit into a desk space of roughly 0.6 m^2 (1.2 m × 0.5 m). The experimental setup presented in this work focuses on the parallelization of individual MEAs while uniting common elements. Replicating the individual MEA cell testing capabilities as a means to scale up is neither cost-effective nor practical in terms of space-occupation due to the fact that the aforementioned equipment employed in the CO_2RR is both expensive and bulky. Therefore, in order to build an effective HTE platform, we pursued miniaturization and cost-effectiveness in the system-level design. In the following subsections, the effect of these two criteria on our HTE design is discussed.

Parallelization of input resources

The HTE system requires multiplication of the input sources such as CO_2, anolyte, and DC power sources. On the cathode side, the CO_2 input to each MEA cell is provided using a single flow controller connected to a flow divider. Instead of adding a mass flow controller on each line, only one standard mass flow controller (Sierra SmarTrak Series 100) is connected to the main CO_2 line. This regulates the total CO_2 amount flowing into the 10 lines that feed the MEA cells. The flow rate along each line can be further tuned using the mechanical valves on the manifold. In order to achieve an even distribution to the MEA cells, pressure regulators are connected to each CO_2 line. The pressure regulators suppress the minor pressure drop differences that may come from the fabrication tolerances and reaction related differences in each cell. The total CO_2 flow rate control *via* the controller can be realized over the pilot module available on the controller or automatically with the master computer if the flow controller is connected to it.

To supply current/voltage to individual MEA cells, a multi-channel potentiostat/galvanostat configuration is required. Owing to the small size of the electrodes (1 cm^2), a current delivering capacity of 1 A for each MEA cell is sufficient to run CO_2RR experiments up to 1 A cm^{-2} current density – well above the current norm in the field.[22,23] The 1 A current capacity applies to each channel and is independent of the currents applied in other channels.

Although this less demanding specification of current capacity may eliminate the need of boosters for some commercially available potentiostat/galvanostat products, the addition of the multi-channel feature makes potentiostats financially infeasible (in addition to their large space requirement). Therefore, typical battery tester units (Landt Instruments – CT3002A) that are utilized in fuel cell evaluation are employed in the HTE platform presented here. The current discharge capacity of 1 A within voltage levels of 0–9 volts for each of the channels satisfied the system needs while utilizing the lab space efficiently. The current and voltage capacity of the system can be easily upgraded by replacing the battery testers with the counterparts or power supplies custom-designed accommodating a variety of electrocatalytic applications without compromising the space or cost. Two battery tester units, each with 8 independent channels, are controlled by the master computer of the HTE platform. The batch or individual commands can be sent easily to the related MEA cell with the software provided by the manufacturer. Considering the high cost of potentiostats, this is a significant benefit to the cost-effectiveness of the system. The utilization of battery testers rather than potentiostats/galvanostats also enabled the building of the HTE setup in a more compact way.

On the anode side, the anolyte used is potassium bicarbonate, which is typically employed in the neutral CO_2-RR. Hence, in the design process, the same manifold type is

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also used on the anode side. A large peristaltic pump (capable of delivering a flow rate of 400 ml min^{-1}) is utilized to circulate the anolyte from a common reservoir to the manifold, which diverts the flow into 10 lines. The anolyte pumped to the MEA cells is then collected back in the common reservoir. A power adapter that is able to tune the voltage applied to the DC motor of the pump is used to control the flow rate of the anolyte. The system could be modified to accommodate cold traps to detect liquid products on cathode outlet streams. To further detect liquid products in the anolyte would be possible by fully decoupling the anolyte streams using separate pumps and reservoirs.

The entire HTE platform employing 10 MEAs covers only 1.35 m^2 (1.5 m × 0.9 m) of a desk footprint as a result of parallelization of input resources whereas the scaling of a conventional MEA testing station by 10 would occupy approximately 6 m^2. For a detailed comparison in terms of cost and lab space coverage, see Table S1.† Fig. 1 shows a schematic of the HTE platform.

New MEA design

The MEA cells employed in the HTE platform have the same flow field features in all 10 units. Each MEA can accommodate a different catalyst and a different local environment depending on the experiment. To keep the design compact, the MEAs are designed to accommodate 1 cm^2 electrodes, but could be scaled up further as needed. Selecting a relatively smaller electrode size also eases the specifications of the potentiostat/galvanostat to be employed in the HTE platform.

The MEA cells are assembled on an 18 cm × 13 cm optical breadboard with clamps. To maintain alignment between the cell plates, four holes are drilled on the corners, where alignment posts pass through.

The MEAs are mounted in the upright (vertical) orientation to save lab bench space and to facilitate the escape of reaction products through bubbles from the active electrode area.

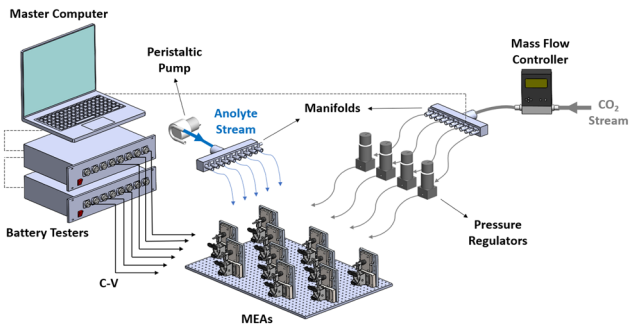

Fig. 1 Schematic of the high-throughput testing system. A mass flow controller and a large peristaltic pump are integrated with manifolds to control CO_2 and anolyte inputs, respectively. The battery testers and the mass flow controller can be controlled by the master computer.

One of the most significant features of our MEA cell design is the ease of assembly. Given the drawbacks of screws used in a typical MEA cell, we designed our cells to be assembled *via* hold-down clamps. Each MEA cell utilizes a hold-down clamp with a locking feature (instead of 4 or more screws) to simply finalize the assembly once all the components are placed between the cell plates. This change in MEA cell design helps us reach the goal of reliability and repeatability which is lacking in conventional MEA cells due to the requirement of sensitive torque tuning in screws and nuts. This problem is eliminated with this design once the applied pressure by the clamp is optimized.

The fast assembly process accelerates the start of the experiments dramatically while preventing the membranes from drying out during the assembly and preparation process. This makes it possible to run multiple MEAs simultaneously. To compare the assembly time of a standard MEA and the MEA presented in this work, eight experts in the field were asked to assemble each type of cell three times, and their efforts are timed. In this experiment, the electrodes and membranes are excluded in the assembly process as well as tubing connections since these are common for the preparation of each MEA type. Table 1 presents the acceleration factor of the assembly processes in a comparison with conventional MEAs. The average acceleration factor is observed to be 8.5 times when compared to the conventional assembly procedures utilizing screws and the maximum observed acceleration was 12.8 times. Fig. 2 presents the statistical overview of the data collected from the assembly time experiment (further details in the ESI†).

The HTE MEA platform protects the electrode structures from extreme compression which may cause mass transport issues as well as membrane damage. To prevent over-compression, a ring shim made of polyether ether ketone (PEEK) is placed between the anode and cathode current collector plates, and sits on a circular protrusion as seen in Fig. 3a. This shim acts as a hard stop and enables the control of the cell compression assuming the electrode structures and membranes have the same features, such as the thickness and porosity of the layers. The effect of different pressure levels on the MEA performance is investigated in the Results section. Fig. 3b illustrates the cross section of the MEA and thicknesses of the shim, the O-ring, and the grooves.

Results and discussion

Experimental results

High-throughput data generation from 10 parallel MEAs was used to study reactor repeatability and catalyst layer parameter sensitivity. The selected catalyst for the repeatability study contains Cu nanopowder (25 nm nanoparticles) and carbon nanoparticles (CNPs). The preparation of the catalyst was repeatable which was the main reason for selecting this catalyst. The details regarding the preparation step for this catalyst are given in the ESI.†

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Table 1 Comparison of the assembly time of the new MEA design and the conventional MEA design

	Conventional MEAs	MEA design presented in this work
Assembly time	68 seconds $\sigma = 21.6^a$	8 seconds $\sigma = 3.3^a$
Acceleration factor	~8.5×	

[a] σ denotes the standard deviation of the timing measurements.

Fig. 4 shows the results of this experiment and highlights the repeatability of the system.

The experiment planned for the parameter sensitivity study employed three variants of the catalyst used in the repeatability experiment using the same materials with different proportions. The major change was the mass ratio of Cu to Nafion, which is a common practice in the field in search of the optimized ratio for the best catalytic performance. The Cu mass in the catalyst ink was doubled, tripled, and quadrupled in these three catalysts, making the Cu : Nafion ratios (4 : 5), (6 : 5) and (8 : 5), respectively. Ten MEAs are divided into three groups to host these three different catalysts in this experiment by 4–3–3. One MEA was subjected to over-compression in assembly in order to monitor how the compression force affects the catalyst performance. Fig. 5 illustrates the gas faradaic efficiency (FE) and cell voltage data collected from the 10 MEAs and Fig. 6 compares the target output products at given current densities for each catalyst ratio.

Both experiments were conducted with a CO_2 flow rate of 200 standard cubic centimetres (sccm) from the main line which is diverted into ten lines and fed to the MEAs. The flow distribution of the MEAs varied between 10–30 sccm. This difference is expected to make no substantial difference in the selectivity of the gas products since the flow rates were above 7 sccm cm^{-2}.[11]

Product analysis of the experiments is limited to gas products, since the motivation is to showcase certain features of the system design only. The output gases were collected using 1 mL injection needles and the content of the gases was analysed by gas chromatography (PerkinElmer Clarus 590). The injections were handled manually in these two experiments. The liquids in both experiments comprise the remaining products, corresponding to 30–35% of the total current towards all products at high current densities (greater than 100 mA cm^{-2}). The catalysts utilized in both experiments resulted in under 0.2% selectivity toward methane.

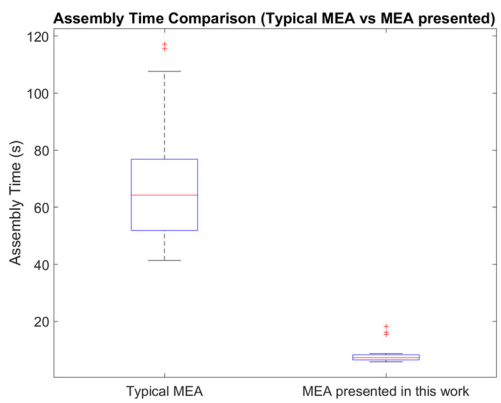

Fig. 2 Statistical overview of the assembly time data for a typical MEA *vs.* the MEA presented in this work.

Discussion

The repeatability experiment results show that a Cu : Nafion ratio of (2 : 5) produces the most ethylene, with an average selectivity of 33% at a current density of 150 mA cm^{-2} within a ±2.5% range as seen in Fig. 4 and 6. This shows that the designed high-throughput experiment platform can deliver reliable and repeatable data to validate catalyst performance. Moreover, with the integration of faster product analysis tools

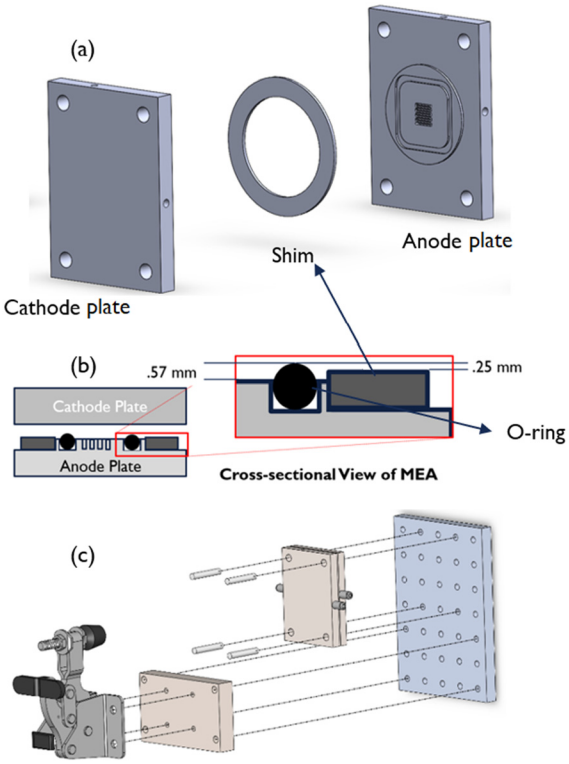

Fig. 3 (a) Major components of our new MEA design: cathode current collector plate, shim, and anode current collector plate. (b) Cross-sectional view of MEAs illustrating the thickness of the O-ring and shim stands. (c) Assembly of elements of a single MEA: clamp, inter-connection part, alignment rods, electrolyzer, and the base.

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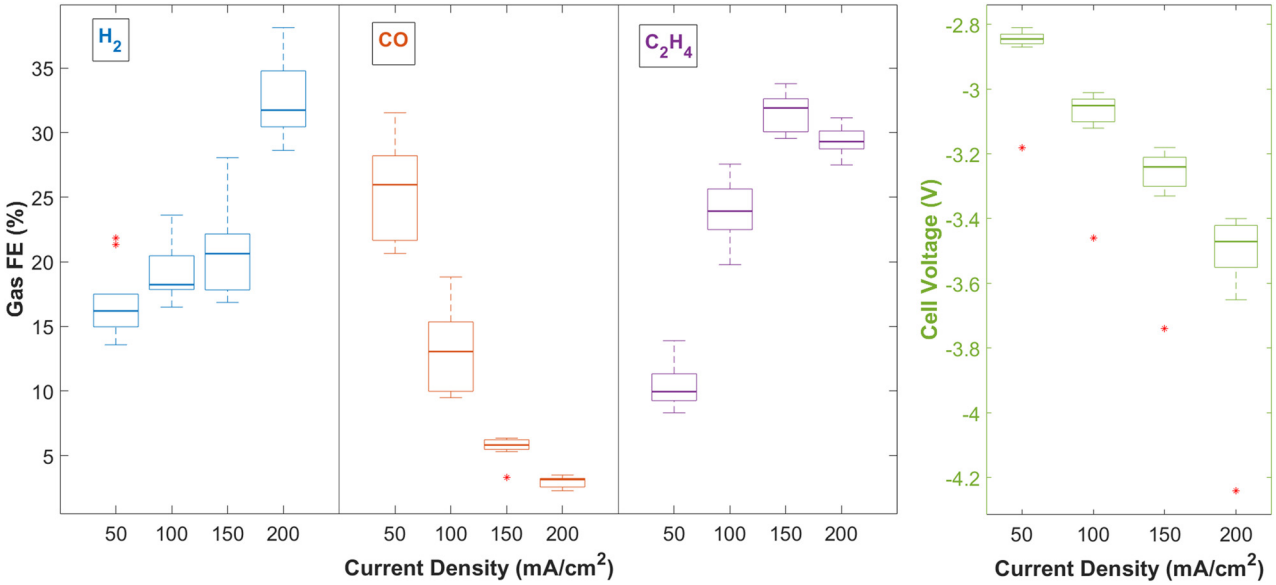

Fig. 4 Overall results of the repeatability experiments running 10 MEAs in parallel. Gas output results are presented over the faradaic efficiency (FE) at the corresponding current density on the left. Full cell voltages recorded from the MEAs during the experiment are presented on the right panel.

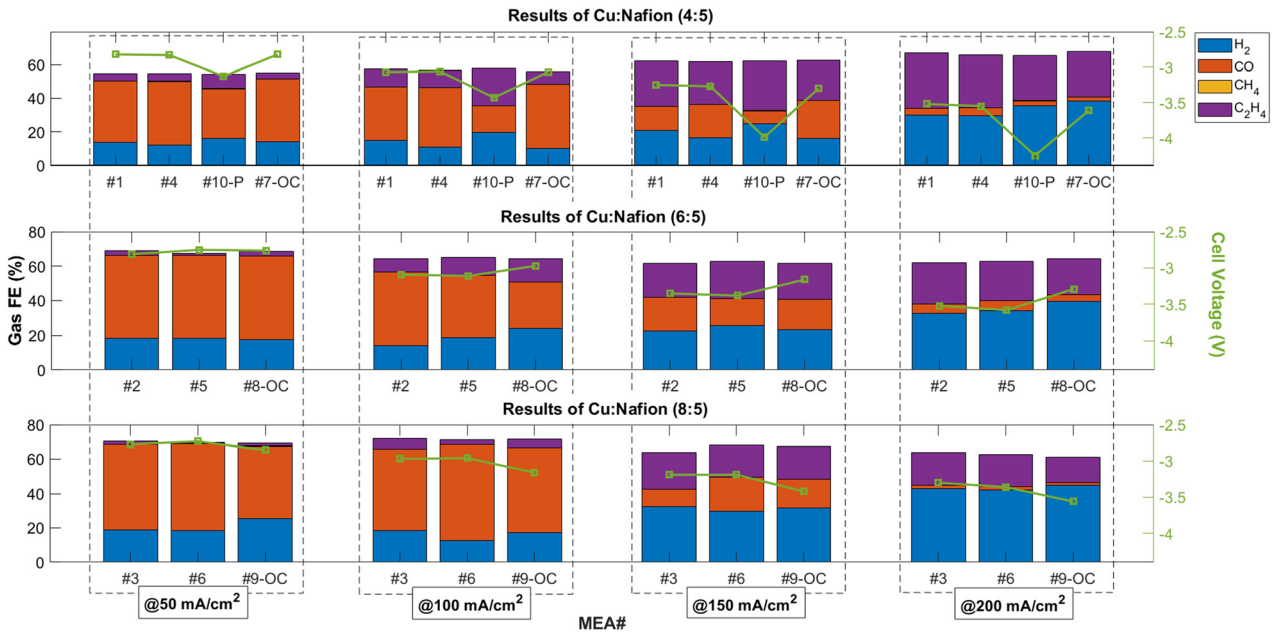

Fig. 5 Comparison of gas phase faradaic efficiencies and full cell voltages of parallel MEAs resulting from the parameter sensitivity study.

such as SIFT-MS, different catalysts can be assessed for specific target products which could enable even greater data generation rates. The deviation of ±2.5% in selectivity may come from inevitable manufacturing imperfections causing slight variations in the electrode or membrane thickness and porosity even though the same materials are employed in this experiment. The variation in selectivity at lower current densities ($j < 100$ mA cm^{-2}) is observed to be higher, which we attribute to the time elapsed from the start of the experiment. This also indicates that a steady state can be reached after 1.5 hours from the start of the experiment.

The outliers (red asterisks) in the cell voltage that can be seen in the right panel in Fig. 4 belong to the prototype MEA which has slightly different current collector features than the other MEAs. These differences are explained in the ESI.† Similar lower cell voltages can also be seen on the line plots in the top panel of Fig. 5 where it represents the cell voltage of MEA#10-P. In order to examine whether

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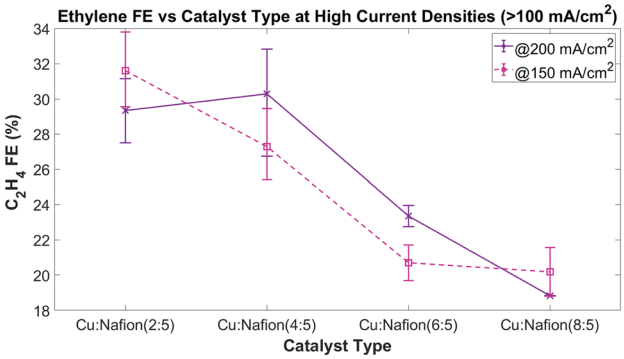

Fig. 6 Comparison of ethylene faradaic efficiency results of different catalysts at high current densities (>100 mA cm^{-2}). The Cu : Nafion (4 : 5) ratio is observed to perform higher at 200 mA cm^{-2} in terms of ethylene selectivity based on the data obtained after two experiments. Note that the data represent the averaged values and extremes at the given current density for the given catalyst.

the results are relevant for scaling up purposes, we also conducted the repeatability experiment with a conventional MEA using a larger (5 cm^2) electrode size (see Fig. S5† for the results). The results are in agreement with what is presented in Fig. 4 for the current density range of 100–200 mA cm^{-2}.

The parameter sensitivity study results presented in Fig. 5 are grouped based on the catalysts utilized in the experiment. Each row belongs to the same catalyst at different current densities and each stacked bar represents the result of that particular MEA at the given current density. The tag 'OC' is used to denote the over-compressed MEAs in the catalyst group and their results are presented in the right most column in each group. In addition, to distinguish the results of the prototype MEA, its results are marked by the tag "-P" in Fig. 5.

The results of the second experiment showed that the highest ethylene selectivity can be obtained using the catalyst with the (4 : 5) Cu : Nafion ratio by 32% at 200 mA cm^{-2} (MEA#1). The over-compressed MEAs indicated the loss of approximately 4–8% of ethylene from its top selectivity and a corresponding increase of hydrogen production of approximately 2–5% when compared to the compression-optimized counterparts. These outcomes confirm that the use of mechanical shims protects the electrodes from breaking and damaging the membranes, which would normally result in failure of the experiment. The drop in ethylene selectivity and increase in hydrogen also indicate mass transport limitations in over-compressed MEAs, which is expected since the thickness of the shim and the O-ring protrusion have tight fabrication tolerances as seen in Fig. 3b in order to fully eliminate the over-compression effects. This drawback can be further minimized by optimizing the shim thickness according to the fabrication results of O-ring grooves for each cell.

Fig. 5 also shows a slight drop in the cell voltages for the over-compressed MEAs, which can be observed in the first and third rows (4 : 5 and 8 : 5 ratio catalysts, respectively) while it is the opposite for the catalyst with a (6 : 5) Cu : Nafion ratio. Although these deviations are in the error range of ±0.2 V as seen in Fig. 4 – right panel, the slight drop in the cell voltage was expected in line with the mass transport issues due to over-compression. The increase in the cell voltage in MEA#8 may also be related with the amount of compression applied to the MEA, which is lower than the other two over-compressed MEAs. The pressure film recordings are presented in the ESI† for further analysis.

Fig. 6 illustrates the variations between the catalyst : ionomer ratios at high current densities ($j > 100$ mA cm^{-2}). Overall, the catalyst containing the (2 : 5) mass ratio of Cu : Nafion performs the best at 150 mA cm^{-2} as seen in Fig. 4 and 6. Additionally, doubling the copper amount increased the ethylene selectivity to 200 mA cm^{-2} when compared to the (2 : 5) ratio catalyst. Further increasing the Cu content, as seen in the (6 : 5) and (8 : 5) catalysts, reduced the ethylene selectivity significantly. All in all, the amount of data collected after two experiments helped in finding the best performing ratio for the given material and supported our findings with statistically meaningful and repeatable data.

The presented high-throughput testing platform enables parallel experiments that can be operated simultaneously and are capable of producing repeatable data thanks to the faster MEA assembly process. Currently, the bottleneck for the acceleration of MEA-based CO_2RR testing is product analysis. Gas chromatography takes approximately 20 minutes to complete a full analysis of an injection. The overall acceleration factor of the HTE platform presented in this work is determined to be 3.4× (full calculation in the ESI,† see Table S4). The overall acceleration can be enhanced significantly by incorporating an in-line, rapid product analysis tool connected to the MEA outputs. This would also eliminate the manual labour required for gas injections and help automate the experimental setup further. The overall system is compatible with established hardware for automated in-line gas sampling (see Design and integration of automated gas sampling in the ESI†).

Conclusions

We presented a high-throughput testing platform that can accommodate 10 parallel electrochemical MEA experiments. The unique design of the MEAs helped accelerate the assembly process which enabled rapid, scalable multi-MEA testing. The presented MEAs also enabled better control of cell compression, eliminating non-uniformity in compression that may come from the user. Repeatability studies showed selectivity to ethylene with a deviation of ±2.5%. Owing to the high data throughput in the 10× system, the optimization of the catalyst ratios for a given electrochemical reaction can be greatly accelerated. In a study of the CO_2RR to ethylene, after only two experiments, an optimized system was found. The acceleration in data collection can be further improved by incorporating this system with a faster product analysis tool, such as SIFT-MS.

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Author contributions

T. K. designed the conceptualization of the systems presented and conducted all experiments, as well as the preparation of the original manuscript. C. P. O helped with the methodology and resources. J. W., J. A., B. S., A. S. and S. A. J. aided with the discussion of the experimental design. Y. C. X. helped with the early-stage experimental work. E. H. S. and D. S. supervised the project. All the authors contributed to the discussion and analysis of data, as well as the editing of the manuscript.

Conflicts of interest

There are no conflicts to declare.

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